

Assignement 1.

Classical Thermodynamics

1. The basic 3 assumptions are:

1. monolayer coverage. Adsorption forms only one layer of molecules on the surface adsorption cannot occur on a site that is already occupied.
2. the surface is uniform and homogenous, all binding sites have the same affinity and are energetically equivalent.
3. No interaction between the molecules.

2.

$$q = \frac{q_{\text{sat}} K(T) P}{1 + K(T) P}$$

a) $\lim_{P \rightarrow \infty} q = \lim_{P \rightarrow \infty} \frac{q_{\text{sat}} K(T) P}{1 + K(T) P} \rightarrow$ we notice the form $\lim_{x \rightarrow \infty} \frac{x}{1+x} = 1$

$$\Rightarrow \boxed{\lim_{P \rightarrow \infty} q = q_{\text{sat}}}$$

b) $\lim_{P \rightarrow 0} q = \lim_{P \rightarrow 0} \frac{q_{\text{sat}} K(T) P}{1 + K(T) P}$ at low P $1 + K(T) P \approx 1$

$$\Rightarrow \text{at low pressure} \quad \boxed{q = q_{\text{sat}} K(T) \cdot P}$$

3. At low pressure $q = \underbrace{q_{\text{sat}} \cdot K(T)}_{K_H \text{ (Henry's coeff.)}} \cdot P$

$$K(T) = k_0 \exp\left(-\frac{\Delta H}{RT}\right) \Rightarrow \boxed{K_H(T) = q_{\text{sat}} k_0 \exp\left(-\frac{\Delta H}{RT}\right)}$$

4. Step 1:

For each temp, a plot q vs P can be made to obtain q_{sat} , k_{263} , k_{293} , k_{313}

Step 2:

$$k(T) = k_0 \exp \left(-\frac{\Delta H}{RT} \right)$$

$$\ln k = \ln k_0 - \frac{\Delta H}{RT} \quad \text{we can plot } \ln k \text{ vs } \frac{1}{T}$$

getting the equation of the plot $y = mx + b$

$$\text{where } m = -\frac{\Delta H}{R} \quad \text{and} \quad b = \ln k_0$$

thus determining ΔH and k_0

Statistical Thermodynamics:

1. Canonical ensemble

It describes a system where N, V and T are constant.

The system can exchange energy with the environment surrounding it.

In the context of adsorption this means that each different state of the system has different energy.

2. $Q = \sum_v e^{-\beta E_v} \quad \beta := \frac{1}{k_B T}$

3. $P_v = \frac{\exp(-\beta E_v)}{Q}$

4.

$$\langle S \rangle = k_B \ln Q + k_B T \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V}$$

$$S = -k_B \sum_v P_v \ln P_v \quad \text{and} \quad P_v = \frac{e^{-\beta E_v}}{Q}$$

$$\Rightarrow S = -k_B \sum_v P_v (-\beta E_v - \ln Q) = k_B \beta \sum_v P_v E_v + k_B \ln Q \sum_v P_v \quad \text{knowing that } \sum_v P_v = 1 \text{ by definition}$$

$$\sum_v P_v E_v = \bar{E}_{\text{tot}}$$

$$\Rightarrow S = k_B \beta \bar{E} + k_B \ln Q \quad \beta := \frac{1}{k_B T}$$

$$= k_B \ln Q + \frac{\bar{E}}{T} \quad \text{using the expression given proofed later on: } \bar{E} = k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V}$$

$\langle S \rangle = k_B \ln Q + k_B T \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V}$

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$$P = k_B T \left(\frac{\partial \ln Q}{\partial V} \right)_{N,T}$$

Starting from: $dE = TdS - PdV$

$$P = T \left(\frac{\partial S}{\partial V} \right)_E - \left(\frac{\partial E}{\partial V} \right)_S$$

$$\left. \begin{aligned} \frac{\partial S}{\partial V} &= k_B \left(\frac{\partial \ln Q}{\partial V} \right) + k_B T \left(\frac{\partial \ln Q}{\partial V} \right) \\ \frac{\partial E}{\partial V} &= k_B T^2 \left(\frac{\partial \ln Q}{\partial V} \right)_{T,N} \end{aligned} \right\} \text{substituted in}$$

$$\Rightarrow P = k_B T \left(\frac{\partial \ln Q}{\partial V} \right)_{N,T} + \cancel{k_B T^2 \left(\frac{\partial \ln Q}{\partial V} \right)_{N,T}} - \cancel{k_B T^2 \left(\frac{\partial \ln Q}{\partial V} \right)_{N,T}}$$

$$E = k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{V,N}$$

$$Q = \sum_{\nu} e^{-\beta E_{\nu}}$$

$$\frac{\partial Q}{\partial \beta} = - \sum_{\nu} E_{\nu} e^{-\beta E_{\nu}} \quad \Rightarrow \quad \frac{\partial \ln Q}{\partial \beta} = \frac{1}{Q} \frac{\partial Q}{\partial \beta} \quad \Rightarrow \quad \frac{\partial \ln Q}{\partial \beta} = - \sum_{\nu} \frac{E_{\nu} e^{-\beta E_{\nu}}}{Q}$$

$$\text{knowing } P_\nu = \frac{e^{-\beta E_\nu}}{Q} \quad \Rightarrow \quad \frac{\partial \ln Q}{\partial \beta} = - \sum_\nu P_\nu E_\nu \quad \text{since } E = \sum_\nu P_\nu E_\nu$$

$$\Rightarrow \quad E = - \left(\frac{\partial \ln Q}{\partial \beta} \right)_{N,V}$$

$$\text{with } \beta = \frac{1}{k_B T} \quad \Rightarrow \quad \frac{\partial \beta}{\partial T} = - \frac{1}{k_B T^2}$$

$$\text{using chain rule } \frac{\partial \ln Q}{\partial T} = \frac{\partial \ln Q}{\partial \beta} \frac{\partial \beta}{\partial T} \quad \text{and substituting } \frac{\partial \beta}{\partial T} \text{ in}$$

$$\Rightarrow \quad \frac{\partial \ln Q}{\partial T} = E \cdot \frac{1}{k_B T^2}$$